Topic 8 – Exchange and transport in animals

Students should:		Maths skills
8.1	Describe the need to transport substances into and out of a range of organisms, including oxygen, carbon dioxide, water, dissolved food molecules, mineral ions and urea	
8.2	Explain the need for exchange surfaces and a transport system in multicellular organisms including the calculation of surface area : volume ratio	1a, 1c 5c
8.3	Explain how alveoli are adapted for gas exchange by diffusion between air in the lungs and blood in capillaries	
8.4B	Describe the factors affecting the rate of diffusion, including surface area, concentration gradient and diffusion distance	
8.5B	Calculate the rate of diffusion using Fick's law: $\text{rate of diffusion} \propto \dfrac{\text{surface area} \times \text{concentration difference}}{\text{thickness of membrane}}$	1a 3a, 3d
8.6	Explain how the structure of the blood is related to its function: a red blood cells (erythrocytes) b white blood cells (phagocytes and lymphocytes) c plasma d platelets	1b 2h
8.7	Explain how the structure of the blood vessels is related to their function	1a
8.8	Explain how the structure of the heart and circulatory system is related to its function, including the role of the major blood vessels, the valves and the relative thickness of chamber walls	
8.9	Describe cellular respiration as an exothermic reaction which occurs continuously in living cells to release energy for metabolic processes, including aerobic and anaerobic respiration	
8.10	Compare the process of aerobic respiration with the process of anaerobic respiration	
8.11	*Core Practical: Investigate the rate of respiration in living organisms*	1a 2a, 2c, 2f 4a, 4c
8.12	Calculate heart rate, stroke volume and cardiac output, using the equation cardiac output = stroke volume × heart rate	1a 2a, 2c 3a 4a, 4c

Use of mathematics

- Demonstrate an understanding of number, size and scale and the quantitative relationship between units (2a and 2h).
- **Calculate with numbers written in standard form (1b).**
- Calculate surface area : volume ratios (1c).
- Plot, draw and interpret appropriate graphs (4a, 4b, 4c and 4d).
- Translate information between numerical and graphical forms (4a).
- Construct and interpret frequency tables and diagrams, bar charts and histograms (2c).
- Extract and interpret information from graphs, charts and tables (2c and 4a).
- Extract and interpret data from graphs, charts, and tables (2c).
- Use percentiles and calculate percentage gain and loss of mass (1c).

Suggested practicals

- Investigate the effect of glucose concentration on the rate of anaerobic respiration in yeast.
- Investigate the short-term effects of exercise on breathing rate and heart rate.